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Weights in the Cohomology of Algebraic Varieties

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1. Let X be a complex algebraic variety (i.e., a separated scheme of finite type over $\mathbb{C}$). We also write X for the usual topological space $X(\mathbb{C})$ underlying X . In this lecture, we describe a remarkable filtration on the rational cohomology groups of X , the *weight filtration*, and we give a collection of results concerning its properties. Its definition will be given in §12. For proofs, we refer to the works cited in the bibliography, where the theorems are often proved in more general settings; working over $\mathbb{C}$ allows us to use simultaneously Hodge theory, actions of Galois groups, and resolution of singularities.

The weight filtration is a finite increasing filtration. We denote it by W . It is also defined in relative situations (or in cohomology with proper support). It depends not only on the topological space X , but also on the way in which X is realized as an algebraic variety. It is compatible with the Künneth isomorphisms (i.e., under the isomorphism $H^*(X \times Y) = H^*(X) \otimes H^*(Y)$, one has

$$W_j(H^*(X \times Y)) = \sum_{j' + j'' = j} W_{j'}(H^*(X)) \otimes W_{j''}(H^*(Y))$$

and is functorial for algebraic morphisms. More precisely, if $f : X \rightarrow Y$ is algebraic, then $f^* : H^*(Y) \rightarrow H^*(X)$ is strictly compatible with the weight filtrations on $H^*(Y)$ and $H^*(X)$: if the class $x \in H^i(X)$ lies in the image of f^* , it has filtration $\leq i$ if and only if it is the image of a class of filtration $\leq i$. More generally, every natural map is strictly compatible with the weight filtrations.

The weight filtration is a discrete invariant; it is invariant under de-

formation of the algebraic structure of X . More precisely, one has the following theorem.

2. THEOREM. *Let $f : X \rightarrow S$ be a morphism. For $t \in S$, let $X_t = f^{-1}(t)$. If the sheaf $R^i f_* \mathbb{Q}$ is locally constant (a local system), then there exists a weight filtration W on $R^i f_* \mathbb{Q}$ by local subsystems, such that the maps $r_t : (R^i f_* \mathbb{Q})_t \rightarrow H^i(X_t)$ are strictly compatible with the weight filtrations and, in particular, wherever r_t is an isomorphism, W induces the weight filtration on $H^i(X_t)$.*

3. Roughly speaking, the weight filtration expresses how the cohomology of X can be built in terms of the cohomology of nonsingular projective varieties. Here are some examples.

EXAMPLE 3.1. If X is proper (= compact, for example projective) and smooth (= nonsingular), then $H^i(X) =_{\text{def}} H^i(X, \mathbb{Q})$ is pure of weight i : $\text{Gr}_j^W(H^i(X)) = 0$ for $i \neq j$. In other words, $W_{i-1}(H^i(X)) = 0$ and $W_i(H^i(X)) = H^i(X)$.

EXAMPLE 3.2. Let X be proper, smooth, connected, of dimension d , and let P be a point of X . Among the cohomology groups with support $H_{\{P\}}^i(X) = H^i(X \bmod (X - \{P\}))$, only the one in degree $2d$ is nonzero, and

$$H_{\{P\}}^{2d}(X) \simeq H^{2d}(X) = \mathbb{Q}. \quad (3.2.1)$$

By our principles, $H_{\{P\}}^{2d}(X)$ is therefore pure of weight $2d$. The inverse of the isomorphism (3.2.1) may be viewed as a Thom–Gysin isomorphism

$$\mathbb{Q} = H^0(P) \simeq H_{\{P\}}^{2d}(X);$$

one observes that it does not preserve weights. The general situation is as follows: for Y a smooth subvariety of pure codimension d in a smooth variety X , the Thom–Gysin isomorphism $H^i(Y) \simeq H_{\{Y\}}^{i+2d}(X)$ carries W^k to W^{k+2d} . Writing (n) for a shift of $2n$ in W ($W(n)_k = W_{k-2n}$), this becomes a filtered isomorphism

$$H^i(Y)(-d) \simeq H_{\{Y\}}^{i+2d}(X).$$

EXAMPLE 3.3. Let X be proper and smooth, and let Y be a smooth (closed) subvariety of pure codimension d . There is an exact sequence

$$\cdots \rightarrow H^i(X) \rightarrow H^i(X - Y) \xrightarrow{\delta} H_{\{Y\}}^{i+1}(X) \rightarrow \cdots$$

By 3.1 and 3.2, $\text{Gr}_j^W(H^i(X - Y)) = 0$ for $j \neq i, i + 1$; W_i is the image of $H^i(X)$.

EXAMPLE 3.4. Let X be proper and smooth except at an isolated singular point P . Suppose that the variety $\tilde{X}$ obtained from X by blowing up P is smooth, and that the exceptional divisor D , the inverse image of P , is smooth: X is obtained from $\tilde{X}$ (proper and smooth) by contracting D (proper and smooth) to a point. The space X has the homotopy type of $[\tilde{X} \cup (\text{a cone with base } D)]$, whose cohomology is computed by Mayer–Vietoris; one obtains an exact sequence

$$\cdots \rightarrow H^{i-1}(D) \xrightarrow{\delta} H^i(X) \rightarrow H^i(\tilde{X}) \oplus H^i(P) \rightarrow \cdots$$

which shows that $\text{Gr}_j^W(H^i(X)) = 0$ for $j \neq i - 1, i$; W_{i-1} is the image of δ . For

$i \neq 0$, it is also the kernel of $H^i(X) \rightarrow H^i(\tilde{X})$, and $\text{Gr}_i^W(H^i(X))$ is the image of $H^i(X)$ in $H^i(\tilde{X})$.

EXAMPLE 3.5. Flag varieties are proper and smooth. The weight filtration on their cohomology is therefore given by 3.1. If G is a connected reductive group, the same result holds for the cohomology of BG [1, III]. This makes it possible to calculate the weight filtration on the cohomology of G , which is related by transgression to that of BG . One finds that $W_i H^i(G) = 0$, and that $W_{i+1} H^i(G)$, zero for even i , is equal to the primitive part of the degree- i cohomology of G [1, III]. If $f : G \rightarrow H$ is an algebraic map between varieties of reductive groups, the inverse image of a primitive rational cohomology class of H is therefore again primitive. For other corollaries, see [1, III].

4. The weight filtration admits a grading in a very strong sense. There exist gradings $\underline{W}$ of the $H^i(X)$ that split W :

$$W_n(H^i(X)) = \bigoplus_{j \leq n} W_j(H^i(X)) \quad (4.1)$$

and that are compatible with the cup product and with higher operations derived from the cup product (Massey products, ...). Since the latter are not defined everywhere, the meaning of “compatible with a grading” must be made precise. The simplest approach is to regard a grading as an action of the group G_m , i.e., an action of $\mathbb{Q}^*$ given by algebraic formulas: to a grading $\underline{W}$ one associates the action in which $\lambda \in \mathbb{Q}^*$ acts on W_j by multiplication by λ^j . “Compatibility” means that $\mathbb{Q}^*$ acts by automorphisms of $H^*(X)$, together with its cohomological grading, cup product, and the higher operations derived from the cup product.

5. For simplicity, suppose that X is connected, and let $\mathcal{M}$ be the minimal model of the rational homotopy type of X in the sense of Sullivan [7]. It is a differential graded algebra in degrees ≥ 0 , connected ($\mathcal{M}^0 = \mathbb{Q}$), free (anti)commutative as a graded algebra, and generated by its indecomposable elements (i.e., $d\mathcal{M} \subset (\mathcal{M}^{>0})^2$). One has $H^*(\mathcal{M}) = H^*(X)$, and if X is simply connected, $\mathcal{M}^{>0}/(\mathcal{M}^{>0})^2 = (\pi_*(X) \otimes \mathbb{Q})^\vee$.

A more precise, and more convenient, statement than the one in §4 is that there exists a grading $\underline{W}$ satisfying (4.1), induced by a grading $\underline{W}$ of $\mathcal{M}$ (in degrees ≥ 0 , the sum of gradings of the $\mathcal{M}^i$), such that d and the product are homogeneous of degree 0; in other words, $\mathbb{Q}^*$ acts by automorphisms of $\mathcal{M}$.

6. The mere existence of W and $\underline{W}$ imposes no restriction on the homotopy type of X : one may always declare that all of $\mathcal{M}$ has weight 0. Moreover, every finite polyhedron has the same homotopy type as an algebraic variety. Indeed, let S be a finite set furnished with a collection $\mathcal{S}$ of subsets (the simplices). Assume that every subset of an element of $\mathcal{S}$ also belongs to $\mathcal{S}$. Identify S with the set of basis vectors of $\mathbb{R}^S$, and for $\sigma \subset S$ let $|\sigma|$ be the simplex spanned by the $s \in \sigma$. We show that $|S| = \bigcup_{\sigma \in \mathcal{S}} |\sigma|$ has the homotopy type of an algebraic variety X . If $|\sigma|_{\mathbb{C}}$ denotes the complex affine space spanned by $\sigma \subset \mathbb{C}^S$, it is enough to take $X = \bigcup_{\sigma \in \mathcal{S}} |\sigma|_{\mathbb{C}}$, of which $|S|$ is a deformation retract. One can veri-

fy that in this example $H^*(X)$ is pure of weight-filtration degree 0.

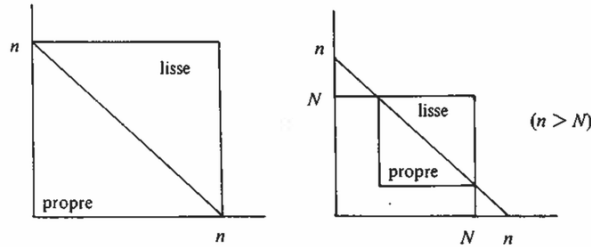
Nevertheless, as soon as the weights are nonzero, constraints appear on the homotopy type of X (cf. §10). The following rules help locate the weights.

7. For a cohomology group H of a given type, each rule will consist in describing a subset $\mathcal{E}$ of $\mathbb{Z} \times \mathbb{Z}$. The weights are controlled by $\mathcal{E}$ in the following sense: if $\mathrm{Gr}_n^W(H) \neq 0$, then there exists $(p, q) \in \mathcal{E}$ with $p + q = n$. This manner of speaking, which may seem artificial here, is not so: information other than the possible weights is controlled by the same region $\mathcal{E}$; cf. §15.

(7.1) $H^n(X)$ is controlled by the square $[0, n] \times [0, n]$.

(7.2) If X is proper, $H^n(X)$ is controlled by the part of this square below the second diagonal, namely $\{(p, q) \in [0, n] \times [0, n] \mid p + q \leq n\}$. If X is smooth, $H^n(X)$ is controlled by the part of this square above that diagonal, namely $\{(p, q) \in [0, n] \times [0, n] \mid p + q \geq n\}$.

(7.3) If $N = \dim X \leq n$, the square $[0, n] \times [0, n]$ may be replaced by the square $[n - N, N] \times [n - N, N]$.



8. Variants. (8.1) The compactly supported cohomology group $H_c^n(X)$ is controlled by the part of the square $[0, n] \times [0, n]$ below the second diagonal. If $N = \dim X \leq n$, this square may again be replaced by $[n - N, N] \times [n - N, N]$. In particular, $H_c^{2N}(X)$ is pure of weight $2N$.

(8.2) The “duality” between the proper and smooth cases follows from Poincaré duality: for X smooth and purely of dimension N ,

$$H^n(X) = \mathrm{Hom}(H_c^{2N-n}(X), H_c^{2N}(X)) = (H_c^{2N-n}(X)^\vee)(-N).$$

This argument extends the results given for smooth X to the case in which X is a “rational homology manifold.” For such an X , the image of $H_c^n(X)$ in $H^n(X)$ is pure of weight n .

(8.3) If X is only assumed normal, it remains true that $H^1(X)$ is controlled by $\{(0, 1), (1, 0), (1, 1)\}$.

9. Let X be a connected variety, $x \in X$, and let $\Pi^{(n)}$ be the largest torsion-free nilpotent quotient of length n of $\pi_1(X, x)$. It is known that finite-index subgroups H of $\Pi^{(n)}$ that are sufficiently small have the following property.

(*) There exists a nilpotent Lie algebra of length n , $(\mathcal{H}, [\ , \])$, with $\mathcal{H}$ a free

$\mathbb{Z}$ -module, such that the Campbell–Hausdorff formula

$$x \circ y = x + y + [x, y]/2 + [x, [x, y]]/12 + [y, [y, x]]/12 + \dots$$

makes $\mathcal{H}$ into a group isomorphic to H .

Moreover, the Lie algebra $\mathcal{L}^{(n)} = \mathcal{H} \otimes \mathbb{Q}$ depends only on $\Pi^{(n)}$, not on H .

10. THEOREM (MORGAN). *If X is normal, $\mathcal{L}^{(n)}$ admits a grading $\underline{W}$ in degrees < 0 , such that $\mathcal{L}^{(n)}$ is generated by elements of degrees -1 and -2 , subject only to relations of degrees $-2, -3$, and -4 , together with the vanishing of n -fold iterated commutators.*

It is more convenient to work with the inverse system $\mathcal{L}^{(\infty)}$ of the $\mathcal{L}^{(n)}$. Defining $H^i(\mathcal{L}^{(\infty)}) = \varinjlim H^i(\mathcal{L}^{(n)})$, one has

$$H^1(\mathcal{L}^{(\infty)}) = H^1(X), \quad H^2(\mathcal{L}^{(\infty)}) \subset H^2(X).$$

Moreover, a grading $\underline{W}$ of $\mathcal{M}$ as in §5 defines compatible gradings of $\mathcal{L}^{(\infty)}$, its cohomology, and that of X . By (8.3) (respectively, (7.1)), $H^1(X)$ has weights 1 and 2 and $H^2(X)$ has weights ≤ 4 . Consequently, $H^1(\mathcal{L}^{(\infty)})$ (the dual of the group $\mathcal{L}^{(\infty)}/[\mathcal{L}^{(\infty)}, \mathcal{L}^{(\infty)}]$ of generators of $\mathcal{L}^{(\infty)}$) has weights 1 and 2, and $H^2(\mathcal{L}^{(\infty)})$, the dual of the group of relations among generators, has weights ≤ 4 . This proves the theorem.

11. For a proper smooth variety, the weight filtration reduces to very little (3.1). Its gradability in the sense of §4 implies the vanishing of all Massey products. Its gradability in the sense of §5 even implies that the rational homotopy type of X can be read off from its cohomology algebra—see [3].

12. Hodge theory. The cohomology group $H^i(X)$ of every complex algebraic variety is endowed with a mixed Hodge structure (W, F) (W is a filtration on $H^i(X)$, and F is a filtration on $H^i(X) \otimes \mathbb{C} = H^i(X, \mathbb{C})$, with W and F satisfying suitable axioms—see [1]). The weight filtration is, by definition, the filtration W .

13. l -adic theory. An algebraic variety X is defined by finitely many polynomial equations. The coefficients of these equations generate a finitely generated subring R of $\mathbb{C}$, with fraction field K , and X is obtained by extension of scalars to $\mathbb{C}$ from a scheme over K , or even over R . The following statements become true after replacing R by $R[1/f]$, with $f \in R$ sufficiently divisible.

(13.1) Let l be a prime number. The group $H^i(X) \otimes \mathbb{Q}_l$ is identified with the l -adic cohomology group $H^i(X, \mathbb{Q}_l)$. The latter is defined in a purely algebraic manner and therefore carries, by transport of structure, an action of $\text{Aut}(\mathbb{C}/K)$.

(13.2) Let $\bar{K}$ be the algebraic closure of K in $\mathbb{C}$. The action factors through $\text{Gal}(\bar{K}/K)$. It is unramified over $R[1/l]$, i.e., it factors through the Galois group of the largest extension $K_{\text{nr}}^{(l)} \subset \bar{K}$ of K unramified over $R[1/l]$.

(13.3) Let $\mathfrak{m}$ be a maximal ideal of $R[1/l]$, and let $N(\mathfrak{m}) = \#R/\mathfrak{m}$ ($R/\mathfrak{m}$ is a finite field). To $\mathfrak{m}$ there corresponds a conjugacy class of Frobenius substitutions $\varphi_{\mathfrak{m}} \in \text{Gal}(K_{\text{nr}}/K)$, whose inverses are the geometric Frobenius substitutions $F_{\mathfrak{m}} = \varphi_{\mathfrak{m}}^{-1}$.

14. THEOREM. *For f sufficiently divisible, one has*

(i) For every l and $\mathfrak{m}$, the eigenvalues α of $F_{\mathfrak{m}}$ (in a suitable finite extension of $\mathbb{Q}_l$) are algebraic integers. For each α , there is an integer $w(\alpha)$ such that all complex conjugates of α have absolute value $N(\mathfrak{m})^{w(\alpha)/2}$.

(ii) Choose a geometric Frobenius $F_{\mathfrak{m}}$, and let ${}_m W_j$ be the sum of the generalized eigenspaces corresponding to eigenvalues α of $F_{\mathfrak{m}}$ such that $w(\alpha) = j$. The filtration by the $\bigoplus_{j \leq i} {}_m W_j$ is independent of $\mathfrak{m}$ and of the choice of $F_{\mathfrak{m}}$; it is rational, and its trace on $H^i(X)$ is independent of l ; it is the weight filtration.

PRINCIPLE OF PROOF. One expresses the cohomology of X , by means of a spectral sequence, in terms of the cohomology of proper smooth varieties, as in [1]. The spectral sequence abuts to a filtration W on $H^*(X)$, which is by definition the weight filtration of Hodge theory. This spectral sequence has an l -adic analogue; the l -adification W_l of W is therefore stable under a Galois group. The Weil conjecture implies that the eigenvalues of $F_{\mathfrak{m}}$ on $\text{Gr}_j^{W_l}$ are algebraic integers whose complex absolute values are $N(\mathfrak{m})^{j/2}$, and the theorem follows.

15. In the cases considered in (7.1), (7.2), and (7.3), the region $\mathcal{E}$ described there controls not only the weights, but also the Hodge numbers and the divisibility of the eigenvalues of the $F_{\mathfrak{m}}$: in (7.1), (7.2), and (7.3), a region $\mathcal{E}$ has been assigned to a group H , and

(a) The nonzero Hodge numbers h^{pq} of the Hodge structures $\text{Gr}_j^W(H)$ satisfy $(p, q) \in \mathcal{E}$.

(b) (Useful only in case (7.3).) If the $(p, q) \in \mathcal{E}$ with $p + q = j$ satisfy $p, q \geq k$, then (still for f sufficiently divisible) the eigenvalues of $F_{\mathfrak{m}}$ with complex absolute value $N(\mathfrak{m})^{j/2}$ are divisible by $N(\mathfrak{m})^k$.

One hopes that this is a special case of a general principle; cf. [4].

16. The following result is clear from the viewpoint of Hodge theory.

(16.1) For odd j , $\dim \text{Gr}_j^W H^i(X)$ is even.

On the other hand, up to now I know how to prove §§ 2 and 5 only by l -adic methods. For §5, the method of §14 yields an l -adic grading; a device of Sullivan then gives the existence of rational gradings of the desired kind.

The functoriality properties of the weight filtration are evident from the l -adic viewpoint (because Galois commutes with everything that can be defined). It is nevertheless useful to prove them from the viewpoint of Hodge theory in order to obtain analogous properties for the filtration F . Morgan has obtained numerous results in this direction—enough to prove by Hodge theory a result slightly weaker than §10.

17. As §5 suggests, a weight filtration also exists on homotopy groups (tensored with $\mathbb{Q}$) under simple-connectivity hypotheses.

It also exists on groups of vanishing cycles (cf. [6]), and I hope that these filtrations will help us to understand the latter better.

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